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Subject

Phase B1: Systems Engineering & Functional Architecture

Author
Reza Heidari (404203211)
Seyed Reza HosseiniTabar (404209706)

Instructor
Dr. Khabdakhsh

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Abstract

This Phase B1 package freezes the system requirements and defines the functional architecture for the 3U precision agriculture CubeSat. The content is derived from the Phase A mission definition and payload trade study, which selected a multispectral pushbroom imager (RGB+NIR), a 650 km sun-synchronous orbit, a 3U bus, a 4.3 kg wet mass estimate, an average sunlit power generation of about 20 W, and an S-band downlink with about 9 dB link margin.

The MBSE lecture emphasizes that requirements should be captured in a requirements diagram with hierarchical decomposition, traceability, use cases, structure (BDD/IBD), behavior (activity and state machine), and parametric analysis. This report follows that approach by providing a mission-to-system traceability matrix, a functional tree, FFBDs, an N2 interface matrix, payload-bus interface requirements.

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1 Introduction

The purpose of Phase B1 is to transform the mission objectives defined during Phase A into a consistent set of system requirements and a preliminary functional architecture. At this stage of the systems engineering process, the emphasis is placed on defining what the system must accomplish before determining how the solution will be physically implemented. The resulting baseline provides a foundation for subsequent design phases and establishes the information required for a successful System Requirements Review (SRR).

The project considered in this report is a 3U CubeSat dedicated to precision agriculture applications. The mission utilizes a multispectral pushbroom imaging payload operating in visible and near-infrared spectral bands to support vegetation health monitoring, NDVI generation, crop stress assessment, and irrigation management. The selected mission architecture includes operation in a 650 km sun-synchronous orbit, onboard data handling and storage capabilities, attitude determination and control functions, and S-band communication links for data transmission to ground users.

This report develops the functional architecture of the spacecraft by establishing traceable relationships between mission requirements, system requirements, operational scenarios, subsystem interfaces, and system behaviors. Functional decomposition, functional flow block diagrams, interface matrices, use cases, structural descriptions, activity descriptions, operational modes, and function allocations are presented to provide a complete system-level representation. Together, these elements create a coherent engineering baseline that supports detailed subsystem design and interface definition activities during later development phases.

2 Mission Requirements

The following source facts from Phase A were used as the frozen baseline:

- Mission type: 3U Earth observation CubeSat for precision agriculture.
- Orbit: 650 km sun-synchronous orbit, inclination 97.8 deg, LTAN 10:30 AM.
- Primary payload: multispectral pushbroom imager (RGB+NIR).
- Primary mission products: vegetation health monitoring, NDVI generation, crop stress detection, irrigation assessment.
- Secondary mission: disaster monitoring, environmental observation, technology demonstration.
- Preliminary budgets: total mass 4.3 kg, average sunlit power about 20 W, science-mode load about 15-16 W, S-band link margin about 9 dB.

2.1 Mission Requirements (MRD)

The mission requirements below are written at the stakeholder level. They state what the mission must accomplish, without yet dictating the detailed implementation.

Table 2 1.Mission Requirements

Mission Req ID	Mission Requirement
MR-01	The mission shall monitor agricultural areas from orbit using Earth observation imagery.
MR-02	The mission shall generate multispectral products suitable for vegetation health analysis and NDVI.
MR-03	The mission shall provide field-scale observations with revisit suitable for periodic agricultural monitoring.
MR-04	The mission shall operate from a 650 km sun-synchronous orbit to maintain stable illumination conditions.

MR-05	The mission shall support secondary disaster-monitoring tasking without degrading the primary agriculture mission.
MR-06	The mission shall be implemented within a 3U CubeSat platform and the mass limit identified in Phase A.
MR-07	The mission shall support a 3-to-5-year operational lifetime.
MR-08	The mission shall deliver usable observation data to ground users through a reliable downlink chain.

2.2 System Requirements (SRD)

The system requirements are derived from the mission requirements and frozen for SRR. Each requirement is measurable and traceable to one or more mission requirements. Verification methods are limited to analysis, inspection, test, and demonstration, consistent with early-phase systems engineering.

Table 2 2.System Requirements

Sys Req ID	System Requirement	Derived from	Verification
SR-01	The spacecraft shall conform to a 3U CubeSat envelope of 10 x 10 x 34 cm and a wet mass not exceeding 4.5 kg.	MR-06	Inspection/ Analysis
SR-02	The spacecraft shall operate in a 650 km sun-synchronous orbit with inclination 97.8 deg and LTAN of 10:30 AM +/- 30 min.	MR-04	Analysis

SR-03	The spacecraft shall carry a multispectral pushbroom payload covering at least Blue, Green, Red, and NIR bands.	MR-01, MR-02	Inspection / Test
SR-04	The payload shall support vegetation-index generation, including NDVI, from Red and NIR measurements.	MR-02	Analysis / Test
SR-05	The imaging system shall provide ground sample distance of 10 m or better at the selected orbit and a swath width of at least 20 km.	MR-03	Analysis
SR-06	During science imaging, spacecraft pointing error shall not exceed 0.2 deg from nadir and attitude rate shall remain below 0.001 deg/s.	MR-01, MR-03	Analysis / Test
SR-07	The attitude control system shall complete detumbling within 3 orbits after deployment.	MR-07	Analysis / Test
SR-08	The spacecraft shall support launch,	MR-05, MR-07	Inspection / Demonstration

	detumbling, commissioning, science, communication, safe, and end-of-life operational states.		
SR-09	The electrical power system shall provide at least 20 W average sunlit generation and maintain a positive margin above science-mode demand.	MR-07, MR-08	Analysis
SR-10	The payload shall consume no more than 6 W average and 9 W peak during imaging.	MR-02, MR-06	Analysis / Test
SR-11	The spacecraft shall downlink payload data in S-band and support command uplink in UHF.	MR-08	Inspection / Test
SR-12	The communications system shall support a payload-data downlink rate of 1 to 2 Mbps with link margin of at least 9 dB.	MR-08	Analysis / Test
SR-13	The onboard data handling subsystem shall provide 32 to 64 GB of storage	MR-05, MR-08	Inspection / Test

	and support priority-based data management.		
SR-14	The spacecraft shall autonomously enter safe mode when low battery voltage, ADCS fault, thermal anomaly, communication loss, or software fault is detected.	MR-07, MR-08	Test / Demonstration
SR-15	The safe mode shall minimize power use, command Sun-pointing attitude, and preserve essential telemetry.	MR-07, MR-08	Analysis / Test
SR-16	The ground segment shall command the spacecraft, receive telemetry, and process observation data for end users.	MR-08	Inspection / Demonstration
SR-17	The mission planning and onboard scheduling functions shall allow secondary disaster imagery to be prioritized over routine agriculture imagery when required.	MR-05	Demonstration
SR-18	The thermal design shall keep payload	MR-07	Analysis / Test

	and avionics within allowable operating limits during eclipse and sunlit phases.		
SR-19	The spacecraft shall maintain electrical and data interfaces between payload and bus through defined ICD-level connectors, power lines, and digital data links.	MR-06, MR-08	Inspection
SR-20	The spacecraft shall timestamp stored observations and housekeeping telemetry for later ground reconstruction and product generation.	MR-08	Inspection / Test

2.3 Requirements Traceability Matrix

The matrix below provides forward and backward traceability between the mission requirements and the frozen system requirements.

Table 2 3.Traceability Matrix

Mission Req	Meaning	Linked System Requirements
MR-01	Agricultural monitoring	SR-03, SR-05, SR-06
MR-02	Multispectral NDVI products	SR-03, SR-04, SR-05
MR-03	Field-scale periodic observations	SR-05, SR-06, SR-17
MR-04	650 km SSO stable illumination	SR-02

MR-05	Secondary disaster monitoring	SR-08, SR-13, SR-17
MR-06	3U and mass limit	SR-01, SR-03, SR-10, SR-19
MR-07	3–5-year lifetime	SR-07, SR-08, SR-09, SR-15, SR-18
MR-08	Deliver data to ground users	SR-11, SR-12, SR-13, SR-16, SR-19, SR-20

3 Functional Architecture

Following the lecture guidance, the system is defined first by what it must do. The functional architecture is therefore written before any hardware detail. It is organized as a functional tree, then refined into FFBDs.

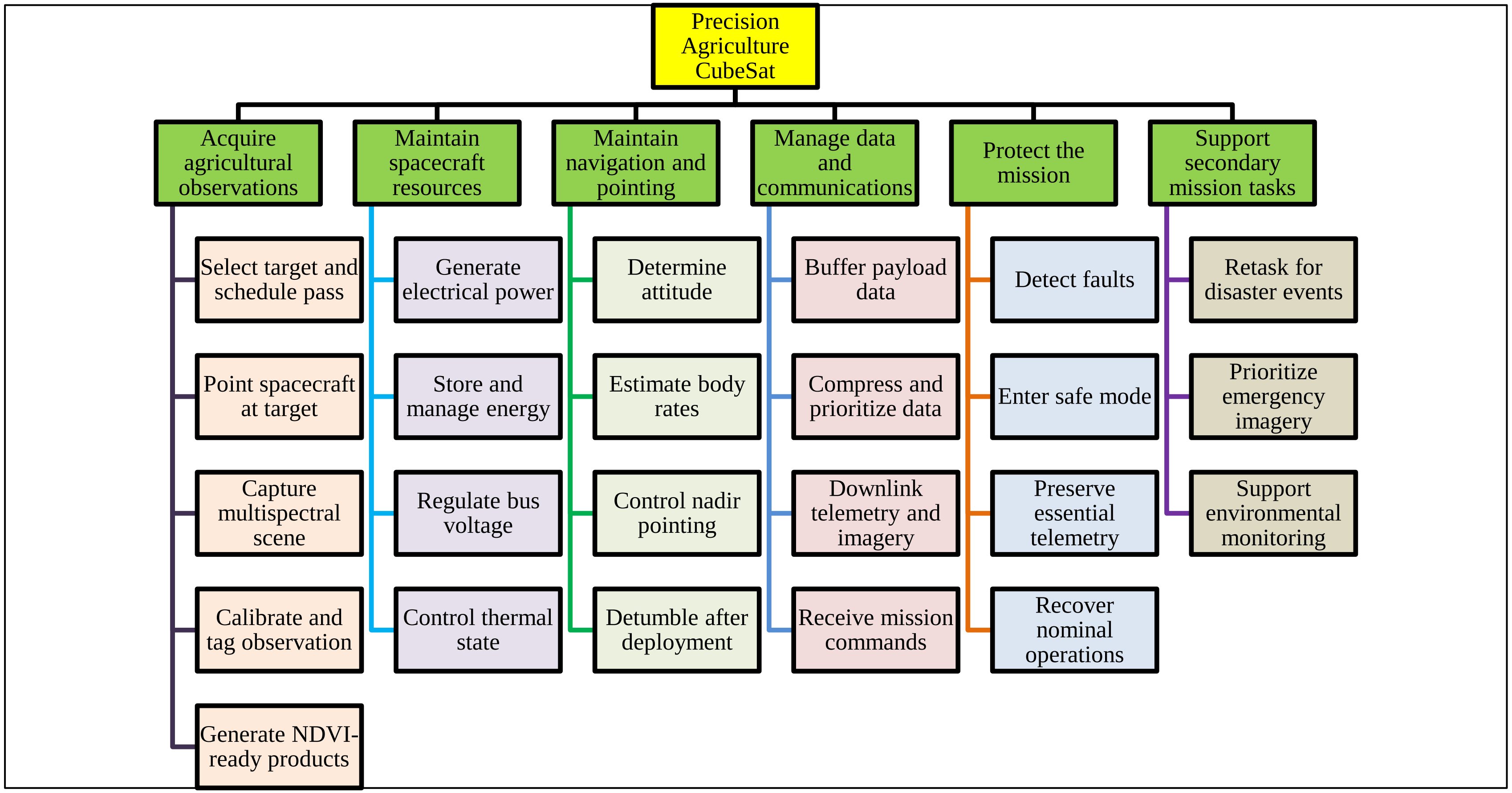

Figure 3 1.Functional Tree

3.1 Functional Flow Block Diagrams

Top-level mission FFBD:

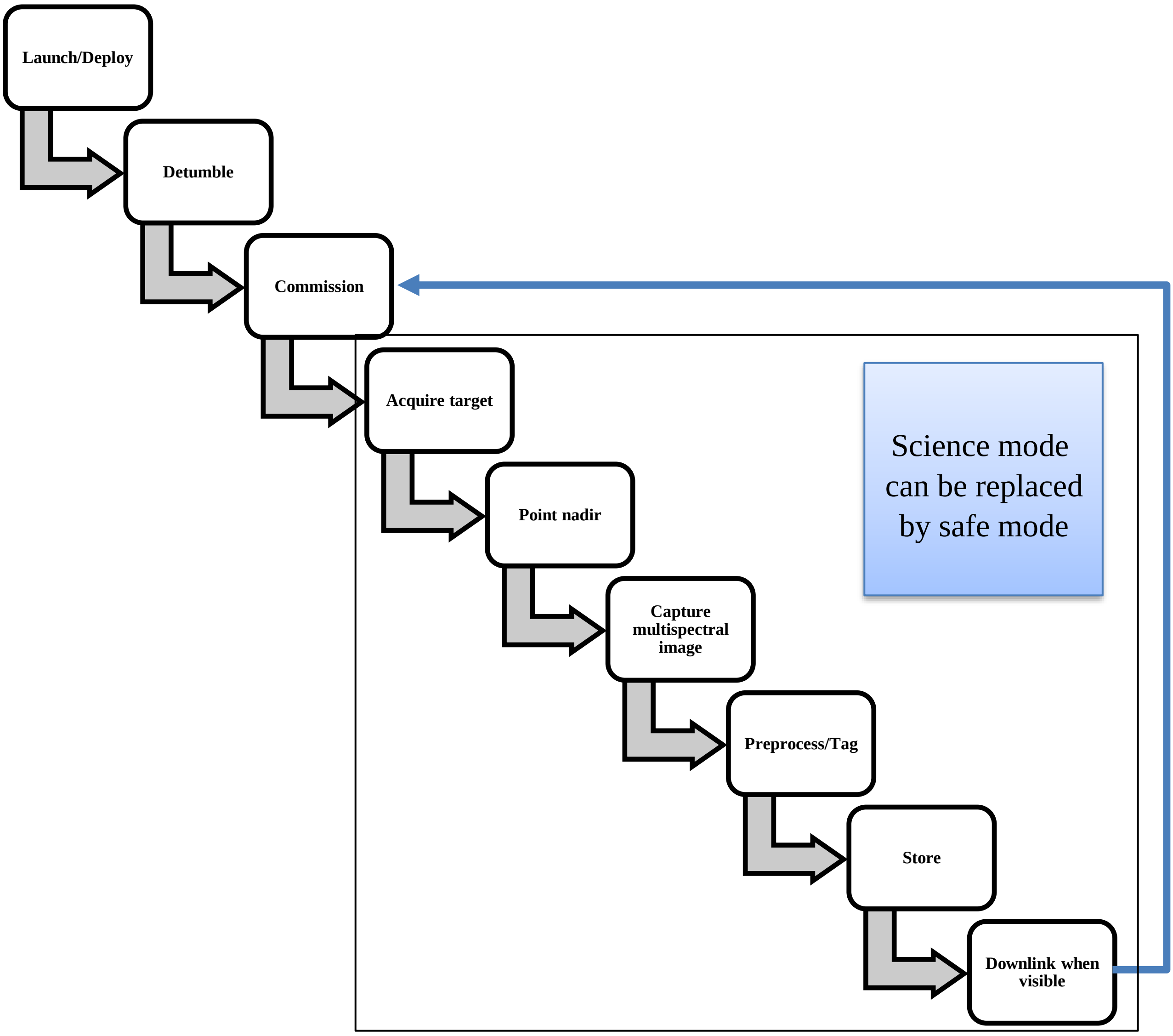

Figure 3 2.FDD

Above, FDD was for top level, now if we want focuse more on detail, we can design FDD for science mode and safe mode.

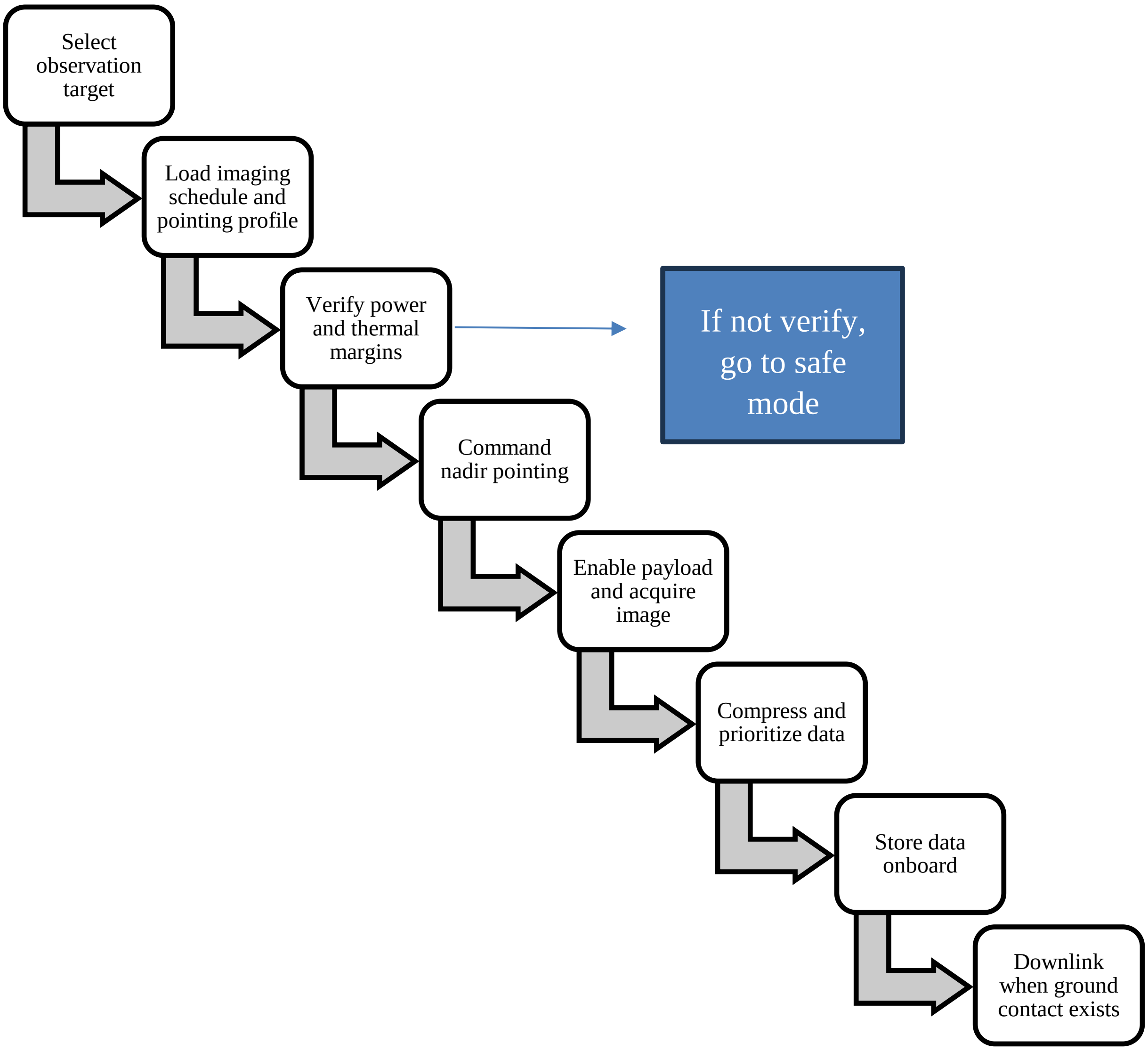

Figure 3 3.FDD for science mode

Science-mode FFBD:

- Select observation target
- Load imaging schedule and pointing profile
- Verify power and thermal margins
- Command nadir pointing
- Enable payload and acquire image
- Compress and prioritize data
- Store data onboard
- Downlink when ground contact exists

Safe-mode FFBD:

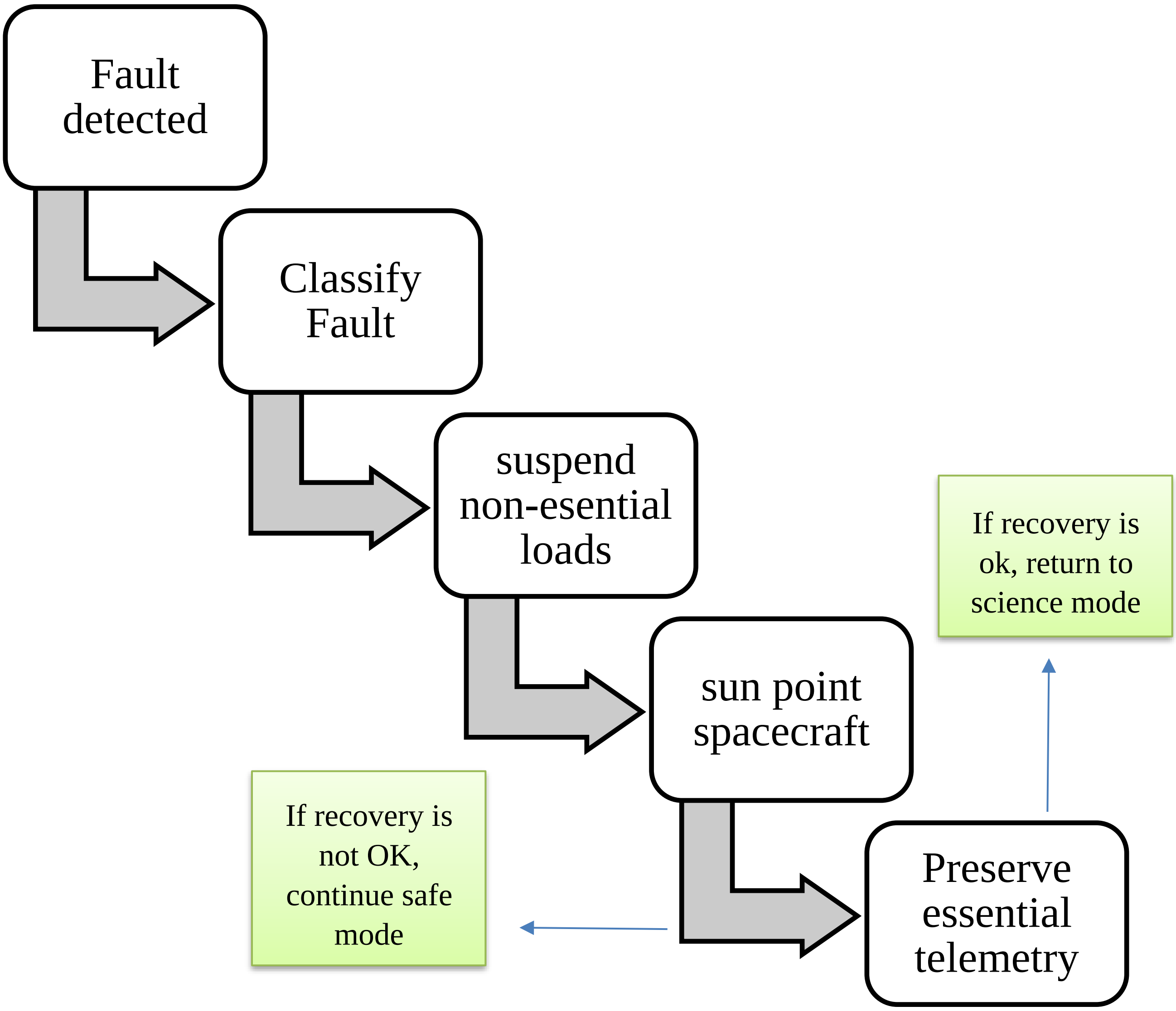

Figure 3 4.FDD for safe mode

3.2 N² Interface Matrix

The N2 matrix identifies the main internal data, power, and command interfaces. It exposes coupling between functions so that interface requirements can be frozen before detailed design.

Table 3 1.Interface Matrix

From\To	PWR	ADCS	PAY	DHM	COM	THM	FDIR	GND
PWR	-	Reg power	Switched power	Bus power/status	TX power	Heater power	Load shed	Power report
ADCS	Att status	-	Pointing enable	Att telemetry	Point cue	Thermal attitude	Fault flags	Attitude report
PAY	Power draw	Boresight req	-	Raw image data	Compressed products	Heat load	Payload fault	Science product request
DHM	Battery status	Att telemetry	Data packets	-	Downlink packets	Memory heat	Storage fault	Stored data / schedules
COM	RF power	Antenna point	Downlink data	Telemetry	-	RF heat	Comm fault	Commands / telemetry
THM	Heater demand	Thermal limits	Payload temp	Temp status	Temp status	-	Thermal alarm	Ground thermal limits
FDIR	Power policy	Safe-point cmd	Payload inhibit	Store priority	Comm inhibit	Heater policy	-	Recovery commands
GND	Mission rules	Pointing plan	Imaging tasking	Data review	Commands	Thermal constraints	Recovery actions	-

interface matrix used to identify and document interactions between the major spacecraft functions and subsystems. Each row represents the source of information, power, commands, or status data, while each column represents the receiving subsystem. The matrix provides a compact representation of all internal interfaces and highlights the dependencies among power, attitude control, payload, data handling, communications, thermal control, fault management, and ground operations. By examining the matrix, engineers can identify which subsystem supplies resources or information to another subsystem and can establish interface requirements before detailed design begins.

The primary purpose of the interface matrix is to support interface control and reduce integration risk. For example, the Payload subsystem provides raw image data to the Data Handling subsystem, the ADCS supplies attitude information to several other functions, and the Power subsystem distributes regulated electrical power throughout the spacecraft. These interactions become inputs for Interface Control Documents

(ICDs) and subsystem specifications. During later design phases, each matrix entry can be expanded into detailed electrical, mechanical, software, and operational interface requirements, ensuring that all subsystem interactions are fully defined, verified, and traceable to mission objectives.

3.3 Payload-Bus Interface Requirements

Table 3 2.Payload-Bus Interface Requirements

Interface Req ID	Payload-Bus Interface Requirement	Verification
PB-01	The payload shall accept switched primary bus power from the EPS and shall operate nominally within the allocated science-mode power budget.	Analysis / Test
PB-02	The payload shall provide a digital data output stream to the onboard computer for storage, compression, and downlink preparation.	Inspection / Test
PB-03	The payload shall present a clearly defined electrical interface for standby, imaging enable, and power-off states.	Inspection
PB-04	The payload shall provide temperature telemetry to the spacecraft housekeeping system.	Test
PB-05	The payload shall remain mechanically and electrically compatible with the 3U volume	Inspection

	allocation and harness routing of the bus.	
PB-06	The payload shall support commissioning checkout and in-orbit calibration sequences commanded by the OBC.	Demonstration / Test

The requirements model is organized hierarchically. Mission requirements contain or derive system requirements, and system requirements further decompose into subsystem-level needs. Each requirement uses a unique ID, natural language text, and a verification method, matching the lecture guidance.

Mission Requirements (MR-01 to MR-08)

- System Requirements (SR-01 to SR-20)
  - Subsystem Requirements (example: ADCS, EPS, Payload, Comms, DHM, Thermal)
    - Component-level implementation requirements during Phase C

3.4 Use Cases

The use-case model describes how external actors interact with the CubeSat system to achieve mission objectives. Use cases provide an operational view of the system and capture the goals that users, operators, and onboard software must accomplish throughout the mission lifecycle. For the precision agriculture CubeSat, the primary actors include the Ground Station, Agricultural Users, Payload Scientists, and the Flight Software. Typical use cases include mission planning, commanding the spacecraft, acquiring multispectral imagery, downloading observation products, responding to anomalies, and recovering from safe-mode events. The use-case model establishes traceability between stakeholder needs and system functions while serving as a foundation for behavioral and operational analyses

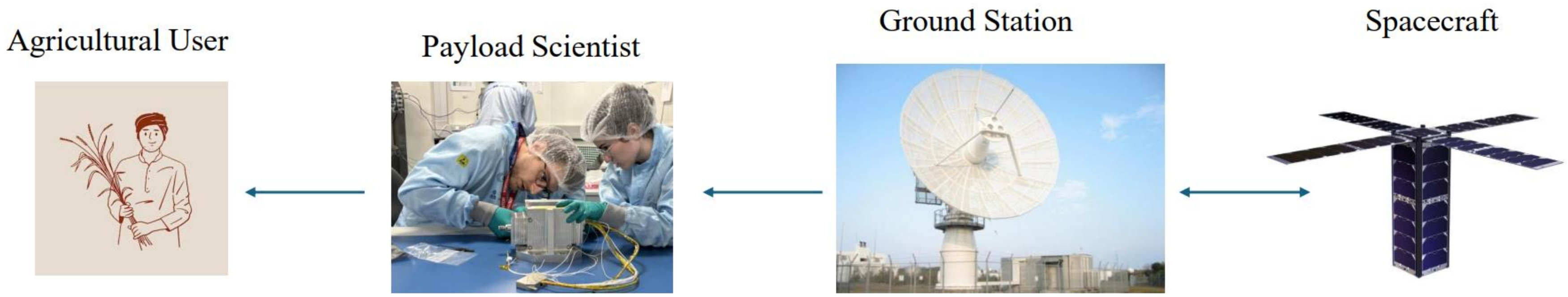

Figure 3 5.Use case

Table 3 3.Use case

Actor	Goal	Use Case
Agricultural User	Upload commands, set mission timeline, trigger safe recovery	Communicate with Ground / Recover Spacecraft
Payload Scientist	Request agricultural or disaster imagery	Perform Mission / Provide Observation Data
Flight Software	Autonomously switch modes based on health and orbit conditions	Mitigate Failures / Manage Modes
Ground Station	Receive telemetry and imagery, return commands, Upload commands, set mission timeline, trigger safe recovery	Communicate with Ground

3.5 Block Definition Diagram

The Block Definition Diagram (BDD) provides a structural representation of the system by identifying the major blocks that compose the CubeSat and defining their relationships. In SysML, the BDD describes the system hierarchy, ownership relationships, and key properties associated with each block. The Spacecraft System is represented as the top-level block and is decomposed into mission management, payload, attitude control, power, communications, onboard computing, thermal control, and structural subsystems. The BDD supports system decomposition, requirement allocation, and architectural consistency by providing a clear view of how the overall mission capability is distributed among the major elements of the spacecraft.

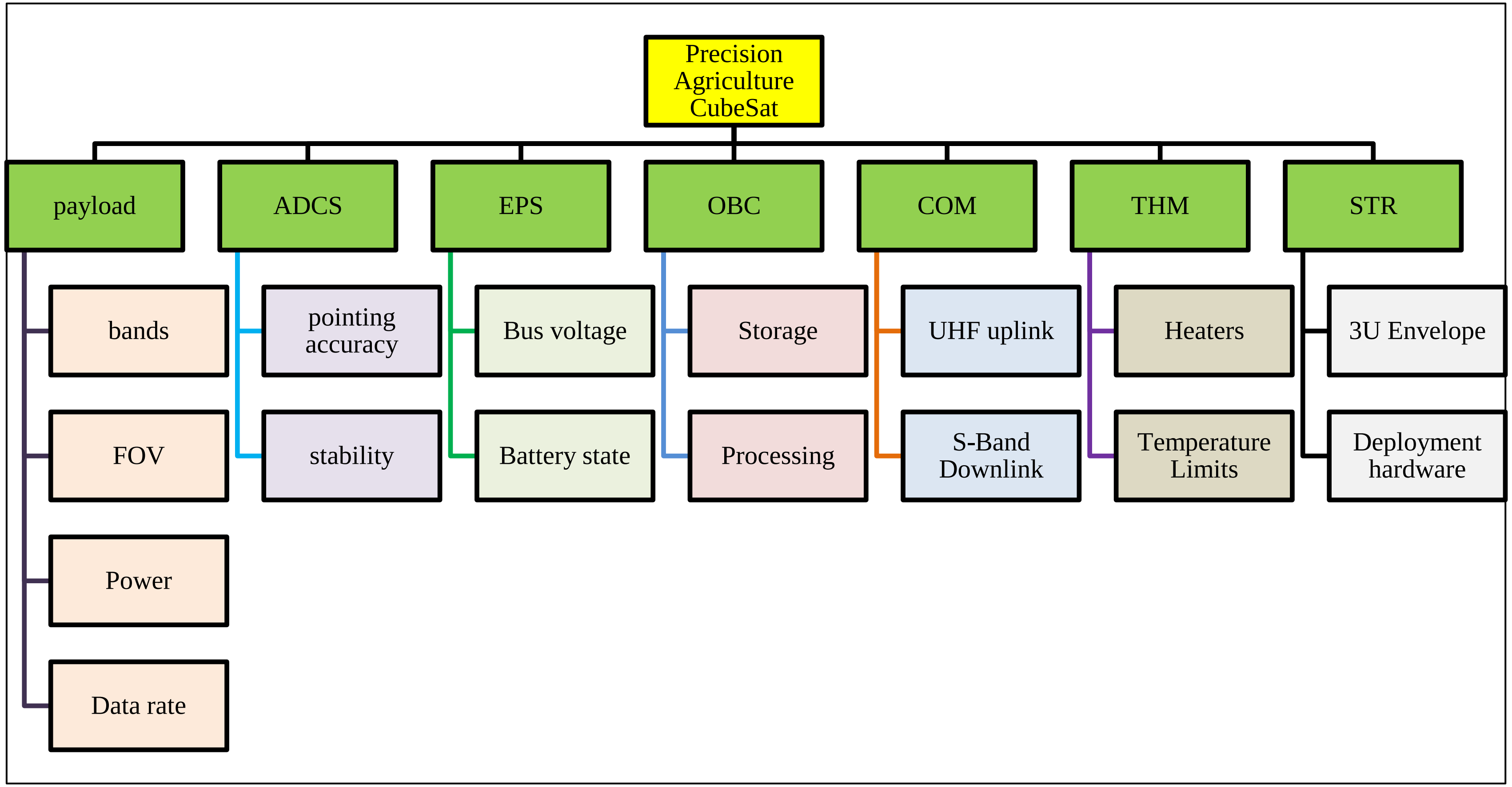

Table 3 4.Block Definition Diagram

Table 3 5.Block Definition Diagram

Block	Role	Value Properties
Spacecraft System	Top-level block	Mass, power, data rate, lifetime
Mission Management	Behavior and mode logic	Schedule, autonomy, mode state
Payload Subsystem	Multispectral pushbroom imaging	Bands, FOV, power, data rate
ADCS Subsystem	Pointing and detumbling	Accuracy, stability, response time
EPS Subsystem	Power generation and regulation	Bus voltage, battery state, margins
OBC / DHM	Onboard control and storage	Memory, processors, packetization
Communications	UHF uplink and S-band downlink	Rates, link margin, antenna gains
Thermal Subsystem	Passive and active thermal control	Temperature limits and heaters
Structure and Mechanisms	3U envelope and deployment hardware	Mass, envelope, interfaces

3.6 Internal Block / Interface Description

The Internal Block Diagram (IBD) describes the internal connections and interfaces between the blocks defined in the BDD. While the BDD identifies the system components, the IBD explains how those components exchange power, commands, telemetry, and mission data. For the CubeSat architecture, the Electrical Power Subsystem distributes regulated power to all onboard elements, the Onboard Computer coordinates command and data exchanges, the Payload provides multispectral imagery to the data-handling functions, and the Communications subsystem manages interactions with the ground segment. The IBD is an essential architectural artifact because it defines the operational relationships among subsystems and serves as a precursor to detailed interface control documentation.

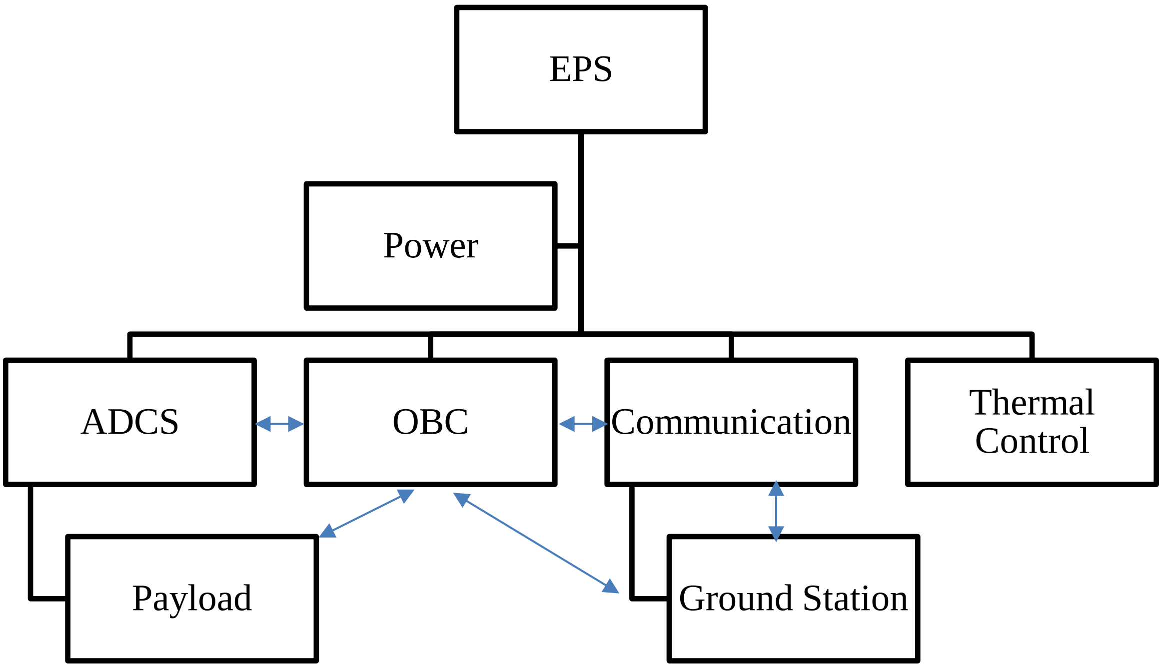

Figure 3 6.IBD

- EPS distributes regulated power to Payload, ADCS, OBC/DHM, Communications, and Thermal control.
- OBC/DHM exchanges commands and telemetry with ADCS, Payload, Communications, and Ground software.
- Payload sends raw multispectral data to OBC/DHM for storage and compression.
- Communications receives command frames from Ground and sends telemetry and products back to Ground.
- ADCS provides pointing state and receives imaging-pointing requests from mission management and payload scheduling.
- Thermal control exchanges temperature telemetry with FDIR and activates heaters when needed.

3.7 Activity / Behavior Description

The activity description captures the sequence of actions required to perform a nominal observation mission. It represents the dynamic behavior of the system and illustrates the flow of information, control, and resources between activities. The operational sequence begins with mission planning and target scheduling, followed by spacecraft health verification, attitude maneuvering, payload activation, image acquisition, onboard data processing, storage, and downlink to the ground station. Activity modeling helps engineers identify operational dependencies, parallel processes, decision points, and resource constraints while providing a foundation for detailed operational procedures and verification planning.

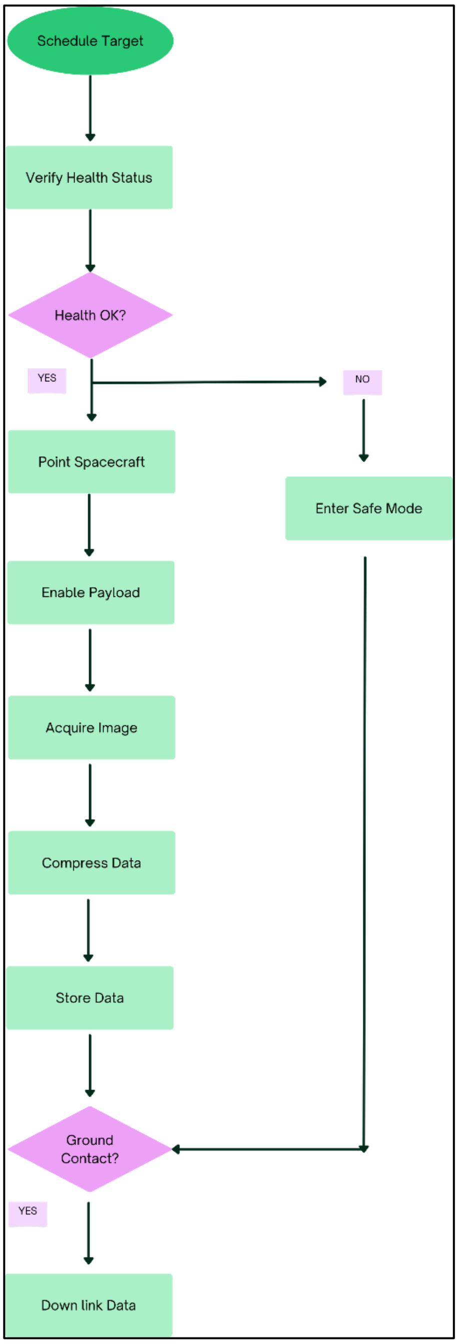

Figure 3 7.Activity Diagram

3.8 State Machine / Operational Modes

The state machine describes the operational modes through which the spacecraft progresses during its mission lifetime and defines the events that trigger transitions between those modes. Following deployment, the spacecraft enters a detumbling state to reduce rotational rates and establish stable attitude conditions. After successful commissioning, the spacecraft transitions into nominal science operations and periodically enters communication mode during ground contacts. In the event of subsystem failures, power shortages, thermal anomalies, or communication losses, the spacecraft automatically transitions into safe mode to protect mission-critical functions. State-machine modeling provides a formal representation of spacecraft behavior and ensures that all operational transitions are clearly defined and verifiable.

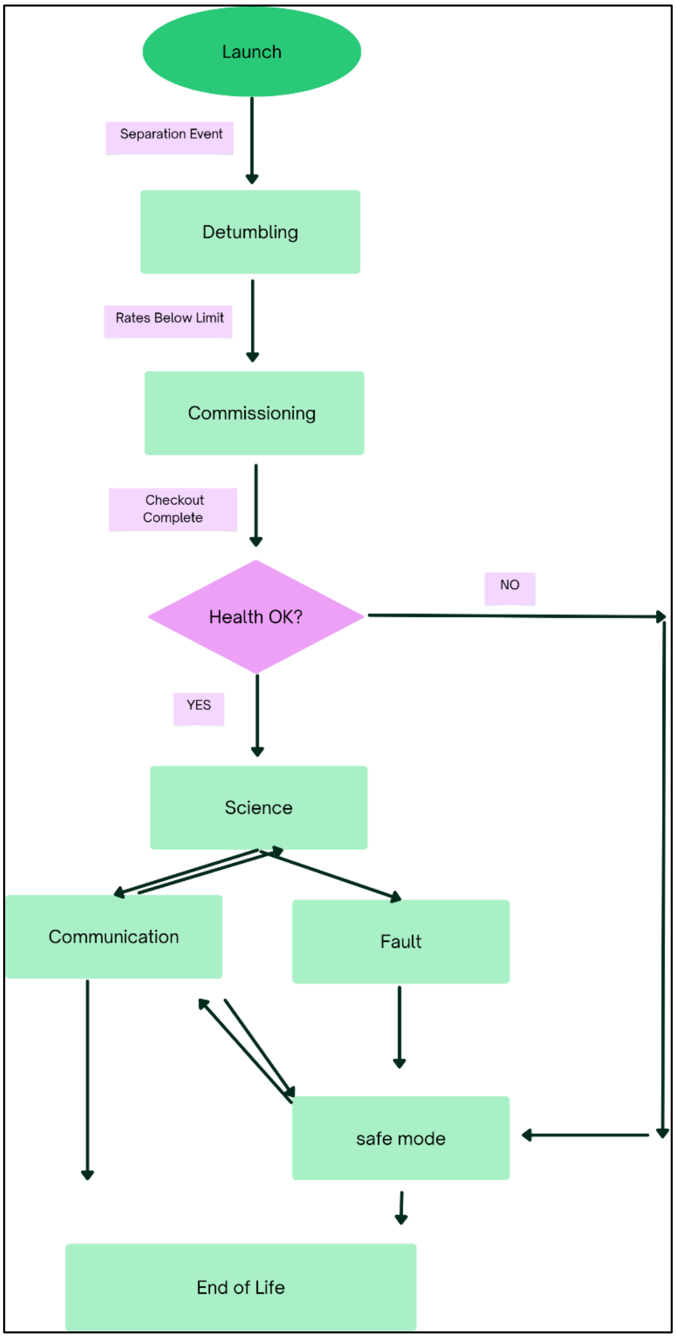

Figure 3 8.Operational Modes

Table 3 6.Operational Modes

State	Purpose	Exit Condition
Launch	Vehicle integration, ascent, deployment protection	Separation from deployer
Detumbling	Reduce rates, acquire Sun, stabilize battery	Angular rate below limit, Sun acquisition
Commissioning	Checkout, calibration, initial imaging tests	Subsystems verified
Science	Scheduled agriculture imaging and data capture	Target list executed
Communication	Telemetry and payload downlink	Ground pass completed
Safe	Fault protection, Sun pointing, essential telemetry	Ground recovery or autonomous exit
End-of-Life	Passivation / disposal	Mission termination

3.9 Allocation of Functions to Subsystems

Function allocation establishes the relationship between system functions and the physical subsystems responsible for performing them. This activity bridges the gap between functional architecture and physical architecture by assigning each required capability to an implementation element. In the precision agriculture CubeSat, power generation and regulation functions are allocated to the Electrical Power Subsystem, attitude determination and control functions are assigned to the ADCS, imaging functions are allocated to the Payload subsystem, data storage and processing functions are performed by the OBC/DHM subsystem, and communications functions are handled by the RF communications subsystem. Proper allocation ensures that all mission requirements are supported by appropriate hardware and software elements while maintaining traceability throughout the system development process.

Table 3 7.Allocation of Functions to Subsystems

Function	Allocated Subsystem
Generate electrical power	EPS / Solar arrays
Regulate bus voltage	EPS
Store and route power	EPS / Battery
Determine attitude	ADCS
Stabilize and detumble	ADCS
Acquire multispectral images	Payload
Store and compress data	OBC / DHM
Transmit telemetry and payload data	Communications
Receive commands	Communications / OBC
Control safe mode and fault response	OBC / FDIR
Maintain thermal limits	Thermal subsystem

4 Conclusion

The Phase B1 systems engineering effort successfully transformed the mission objectives of the 3U Precision Agriculture CubeSat into a structured and traceable system architecture. Mission requirements were refined into measurable system requirements, and complete traceability was established between stakeholder needs and system-level capabilities. The functional architecture was developed through functional decomposition, Functional Flow Block Diagrams (FFBDs), interface definitions, and subsystem allocations, providing a comprehensive description of how the spacecraft will perform agricultural monitoring, multispectral imaging, onboard data management, and communication with ground users. The resulting architecture ensures that all major mission objectives are supported by clearly defined functions, interfaces, and operational behaviors.

In addition, the report applied Model-Based Systems Engineering (MBSE) concepts by incorporating use cases, structural descriptions, behavioral models, operational modes, and interface analyses. These artifacts improve system understanding, reduce integration risks, and establish a solid foundation for subsequent design activities. The completed Phase B1 package provides an SRR-ready baseline that can be used to support preliminary subsystem design, interface control development, performance analysis, and verification planning during the next project phase. As the project progresses toward detailed design, the models and requirements defined in this report will serve as the primary reference for maintaining consistency, traceability, and technical maturity throughout the CubeSat development process.

References

[1] SSD-2026-02-Model-Based System Engineering

[2] PROJ-SYS-RPT-002